Laboratory Work 3

Course

Microcontroller Systems

Title

USART Communication with TX/RX Handling and LED Control

Objective

To extend the previous project with USART packet transmission on button press and byte reception for LED control according to variant rules.

Variant

Variant: **5**

RX command mapping: - `4` -> LED OFF - `5` -> LED ON

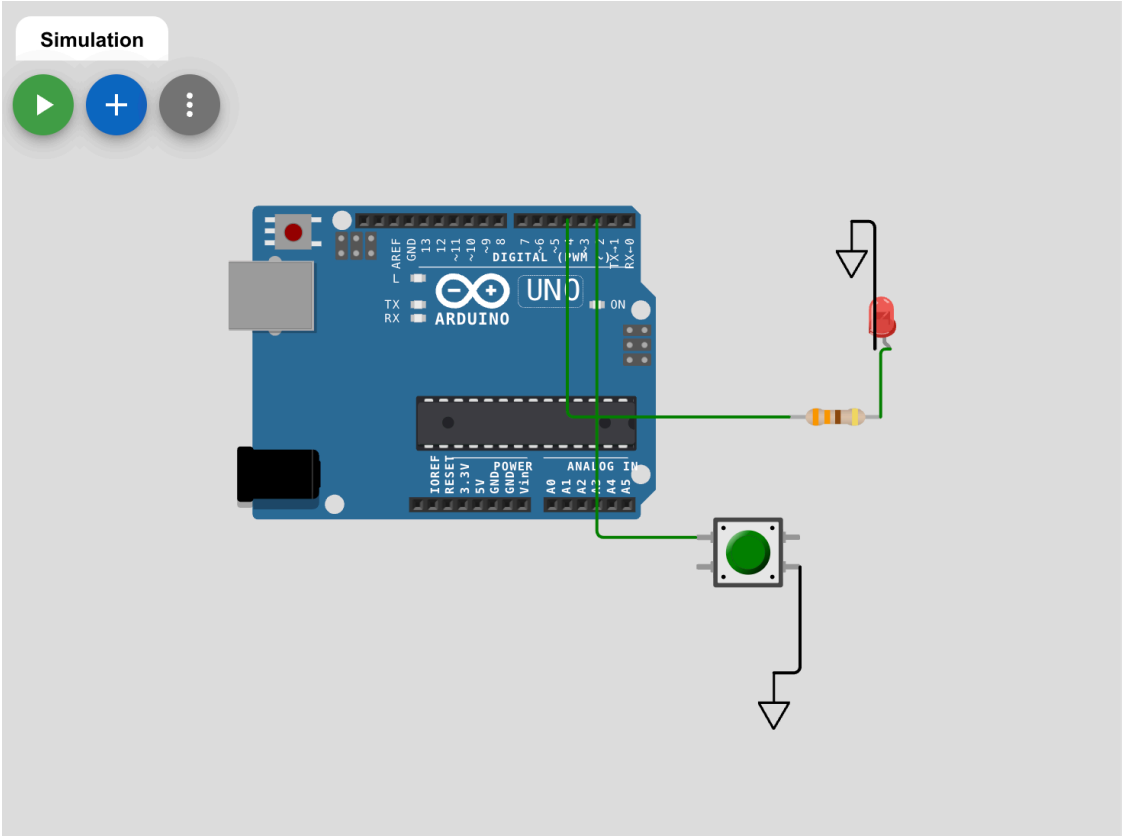
Task Requirements and Implementation

1. **Send function for byte packet**: implemented as `sendPacketIRQStyle()`.
2. **Transmit flow with TXE-like behavior**: byte-by-byte transmission is performed in software loop (functional simulation analogue).
3. **Button-triggered sending**: pushbutton on digital pin `2` launches packet transmission.
4. **RX processing**: received byte is checked and LED on digital pin `4` is switched ON/OFF for variant values.

Circuit / Simulation Setup

Functional simulation setup was implemented in Wokwi: - Controller board: Arduino Uno (used as simulation analogue due stable UART terminal support in Wokwi). - Pushbutton: `D2` to GND (active low with internal pull-up). - LED line: `D4 –> R(330 Ohm) –> LED –> GND`. - UART terminal: built-in Serial Monitor (`9600` baud).

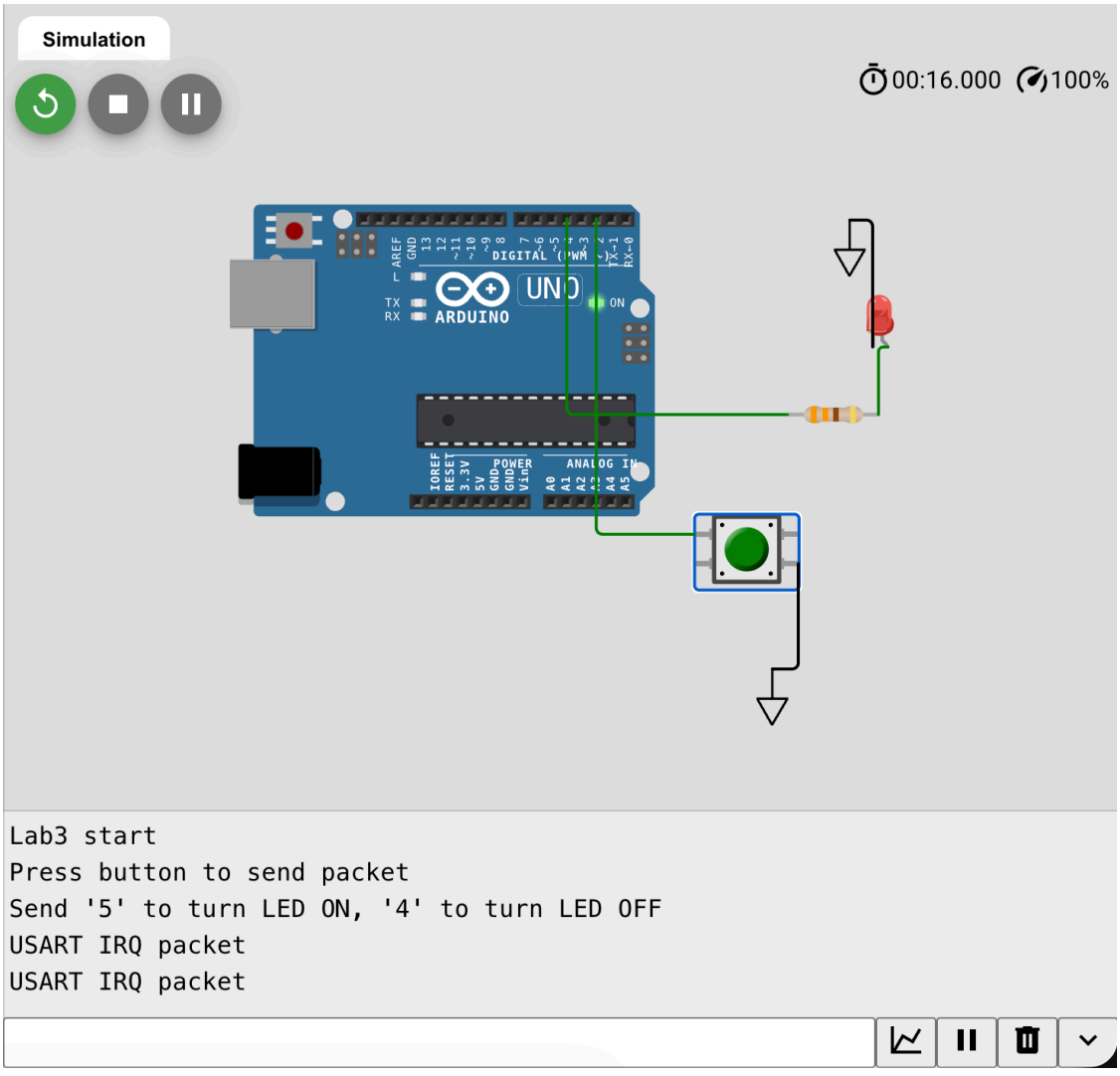
Schematic screenshot:



Lab 3 scheme

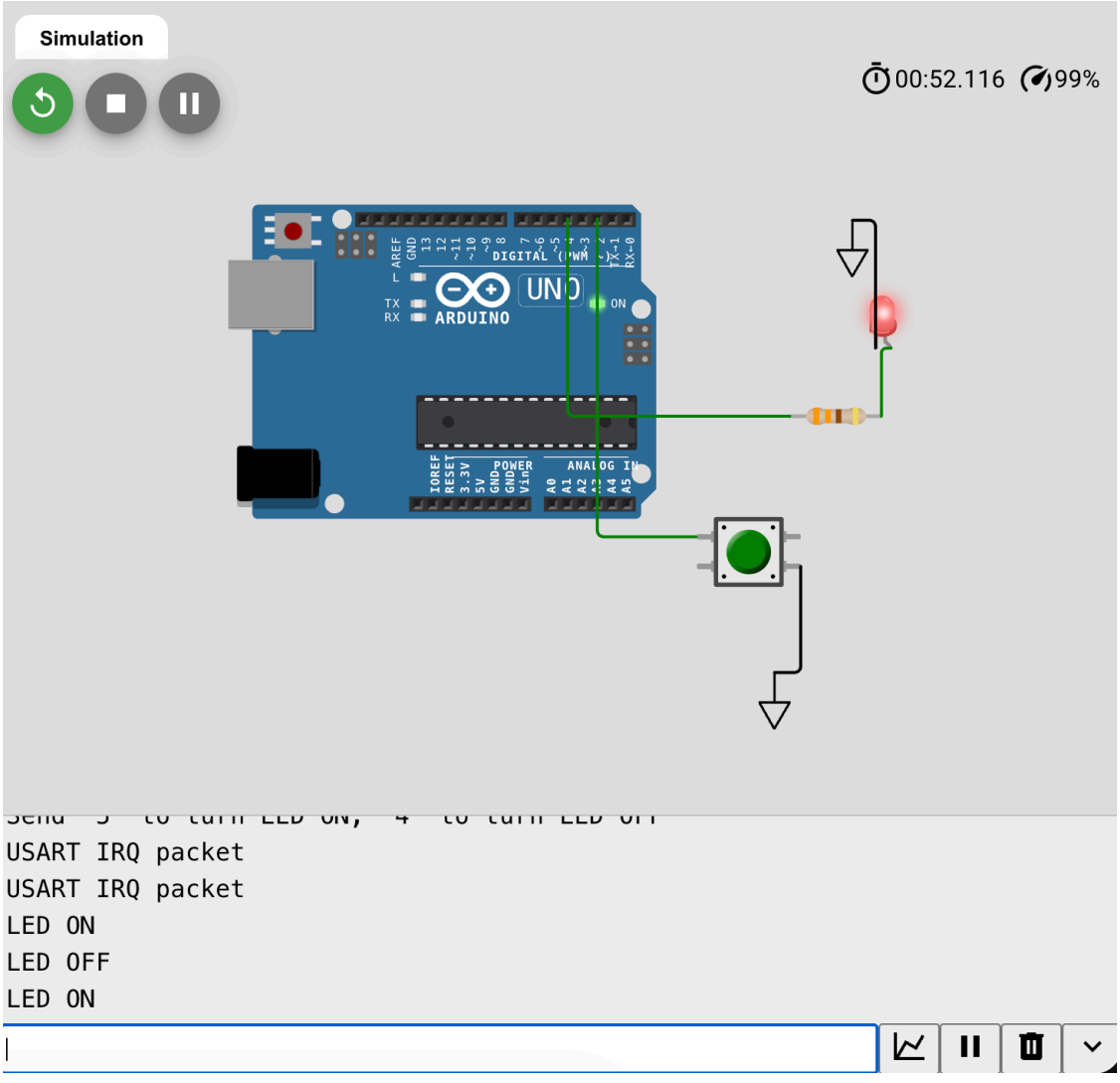
Runtime Verification

TX packet after button press:



Lab 3 UART TX

RX command processing and LED control:



Lab 3 UART RX and LED

Observed behavior: 1. On startup, Serial terminal prints initialization text. 2. Button press sends packet `USART IRQ packet`. 3. Entering `5` in terminal turns LED ON. 4. Entering `4` in terminal turns LED OFF.

Calculations

USART baud in simulation: - Configured baud rate: `9600` bps. - Terminal baud rate: `9600` bps. - Baud mismatch in this setup: `0%`.

Variant logic check: - Task table for variant 5: `4-off, 5-on`. - Implemented mapping matches the task.

Code Listing (`Wokwi_Lab3_sketch.ino`)

```
const int BTN_SEND = 2;
const int LED_CTRL = 4;

const char CMD_OFF = '4';
const char CMD_ON  = '5';

bool btnPrev = false;

void setup() {
  pinMode(BTN_SEND, INPUT_PULLUP);
  pinMode(LED_CTRL, OUTPUT);
  digitalWrite(LED_CTRL, LOW);

  Serial.begin(9600);
  delay(500);
  Serial.println("Lab3 start");
  Serial.println("Press button to send packet");
  Serial.println("Send '5' to turn LED ON, '4' to turn LED OFF");
}

void sendPacketIRQStyle(const char *s) {
  while (*s) {
    Serial.write(*s++);
    delay(1);
  }
}

void loop() {
  bool btnNow = (digitalRead(BTN_SEND) == LOW);

  if (btnNow && !btnPrev) {
    sendPacketIRQStyle("USART IRQ packet\\r\\n");
  }
  btnPrev = btnNow;

  while (Serial.available() > 0) {
```

```
    char rx = (char)Serial.read();

    if (rx == CMD_ON) {
      digitalWrite(LED_CTRL, HIGH);
      Serial.println("LED ON");
    } else if (rx == CMD_OFF) {
      digitalWrite(LED_CTRL, LOW);
      Serial.println("LED OFF");
    }
  }
}
```

Additional microcontroller implementation file (STM32 source used in project structure): - `Lab3_main.c`

Conclusion

Laboratory work 3 was completed. Packet transmission on button press and receive-based LED control were implemented and verified. Variant 5 conditions (`4/5`) are fully satisfied.